KENYA · SCHOOL RESOURCE MAPPER

Shule Atlas

Project documentation

November 2026

A working map of school resources across Kenya.

Shule Atlas places primary and secondary schools on a map, scores each one against a fixed set of planning lines, and filters the set by the gaps a planner is trying to close. This document describes what the app does, how the index is calculated, what is stored, and how to run the project.

Catalogue	85 named schools, all 47 counties
Bands in the catalogue	33 steady · 26 strained · 26 critical
Index	0–100, enrollment-weighted at county level
Edits	Saved in the browser only (localStorage)
Stack	TanStack Start, React, Leaflet, Zustand, Recharts

1. Purpose

Shule Atlas is a planning desk for school resources. It is meant for someone comparing schools inside a county, or comparing counties, on the things that show up in a capitation or infrastructure conversation: teachers, classrooms, books, toilets, desks, power, water, and a short list of learning facilities.

It is not a school inspectorate, an admissions list, or a substitute for the National Education Management Information System. Names in the catalogue are real schools. The counts, ratios, coordinates, and facility flags are composites built so the map, the filters, and the charts can be used. Treat every figure as illustrative until it has been checked against the county education office or NEMIS.

2. Using the app

The screen is one workspace with three views. On a wide display the map stays beside the list. On a phone, Map, Schools, Gaps, and Counties are separate tabs.

Map

- Each school is a pin. Green is steady (index 75–100), amber is strained (50–74), and red is critical (0–49).
- Selecting a school in the list flies the map to that pin and opens the profile.
- Fit schools frames whatever the current filters leave on the map.
- While logging a school, Drop a pin on the map writes latitude and longitude into the form.
- Tiles are CARTO Positron, built on OpenStreetMap. The map needs a network connection.

Directory and filters

Search matches school name, county, and sub-county. Three chips show how many schools in the current filter sit in each band; choosing a chip sets the band filter. The filter panel adds:

- County — any of the 47 counties, or all.
- Level — primary or secondary.
- Ownership — public, private, or faith.
- Gap — a specific shortfall, such as no electricity, no safe water, a teacher shortfall, crowded classrooms, textbooks, toilets, desks, no science lab, no computer lab, no library, no internet, or no school meals.
- Sort — greatest need (lowest index first), strongest index, name, or enrollment.

The header counts update with the filter: schools shown, learners in those schools, and counties represented.

School profile

A profile shows the resource index, the band, and each planning line as a bar. Ratios are written in planner language (“64.6 learners per teacher · line 35”), not as abstract scores. Facilities are a checklist. Anything short of the line is listed again under Shortfalls. Notes travel with the school.

A catalogue school can be updated. The change is stored as an override and can be reverted, which restores the shipped profile. A school logged in the browser can be updated or removed. Removing it does not affect the catalogue.

Gaps

Gaps lists every school in the current filter that misses at least one planning line, lowest index first. Each row names the shortfalls in the same language as the profile. If the filter is already tight and every remaining school meets every line, the view says so instead of showing an empty table with no explanation.

Counties

Counties rolls the filtered schools up to county. The county index is the enrollment-weighted average of school indexes, then banded with the same 75 and 50 cut points. A bar chart and a sortable table show the index, the number of schools, enrollment, and how many of those schools are critical. Weighting by enrollment stops a small school from moving a county average as much as a large one.

Log a school and export

Log school opens a form: name, county, sub-county, level, ownership, coordinates, learners, teachers, classrooms, books per learner, toilets, desks, seven facility checkboxes, and notes. The form rejects a blank name and rejects coordinates outside Kenya's rough bounding box. Save on this browser adds the school to the map immediately and selects it.

Export view downloads a CSV of the schools in the current filter, not the whole catalogue. The file is named shule-atlas.csv. Columns are listed in section 5.

3. The resource index

Every school receives an index from 0 to 100. The index is a weighted average of the parts that apply to that school. A part scores 1 when it meets the planning line and 0 when it misses the line completely. Ratios in between are linear.

Planning lines

Measure	Line	Direction
Learners per teacher, primary	40	Lower is better
Learners per teacher, secondary	35	Lower is better
Learners per classroom	45	Lower is better
Core textbooks per learner	1.00	Higher is better
Learners per toilet	30	Lower is better
Desks per learner	0.5	Higher is better
Electricity, safe water, internet, library, computer lab, meals	Present	On scores 1, off scores 0
Science lab	Present	Scored for secondary only

Planning lines used by scoreSchool in src/lib/model.ts. They are app conventions, not a citation of a circular.

How a ratio is scored

For a "lower is better" ratio, the score is 1 when the actual figure is at or inside the line. Past the line it falls as 1 − (actual − line) / line, and it is floored at 0. A school at twice the line therefore scores 0 on that part. Example: a secondary school at 70 learners per teacher against a line of 35 scores 0 on teachers.

For a "higher is better" ratio, the score is actual / line, capped at 1. Half a book per learner against a line of 1.00 scores 0.5. Two books per learner still scores 1; extra books do not raise the index further.

A school with zero toilets scores 0 on toilets and is described as "No toilets recorded". Enrollment divided by zero is never shown.

Weights

Weights sum to 100 when every part applies. The science lab (weight 5) applies only to secondary schools. For primary schools that weight is left out of both the numerator and the denominator, so a primary school is not

punished for having no lab and the index is still on a 0–100 scale.

Part	Weight	Kind
Teachers	18	Ratio
Classrooms	14	Ratio
Textbooks	14	Ratio
Toilets	14	Ratio
Electricity	8	Facility
Safe water	8	Facility
Science lab (secondary only)	5	Facility
Desks	4	Ratio
Internet	4	Facility
Library	4	Facility
Computer lab	4	Facility
School meals	3	Facility

Applicable weights are renormalised. Primary schools are scored out of 95 points, then scaled to 100.

Bands, gaps, and the county rollup

Band	Index	Meaning in the app
Steady	75–100	Meets most lines. Pins and chips are green.
Strained	50–74	Mixed. Several lines are missed or only partly met.
Critical	0–49	Far from the lines. Listed first under greatest need.

A gap is any applicable part whose score is below 1. Meeting a line exactly does not produce a gap. The Gaps view and the gap filter both use this list.

The county index is round(sum of index × enrollment / sum of enrollment) for the schools currently in view. It is then banded with the same cut points. Sorting the county table by index therefore ranks counties by that weighted figure, not by the unweighted average of the schools drawn on the map.

4. What is stored

The catalogue ships in source, in src/data/catalogue.ts. It is not fetched from a server. Each record has an id, name, county, sub-county, level, ownership, latitude, longitude, enrollment, teachers, classrooms, textbooks per pupil, toilets, desks, seven resource flags, notes, and an origin of catalogue.

Anything a person types is kept in the browser under the localStorage key shule-atlas-v1. The store keeps two lists: schools added on this browser, and overrides of catalogue schools. On load, each catalogue school is replaced by its override if one exists, then the added schools are appended. The server render does not read localStorage; the client applies it after hydration so the first paint matches the catalogue and the saved edits appear immediately after.

Clearing site data for the origin deletes every field log and every local edit. There is no account, and nothing typed into Log school is sent to a database by the mapper.

5. CSV export

Export view writes the filtered, scored schools. Boolean facilities are true or false. Ratios are rounded to one decimal place. An empty learners-per-toilet cell means no toilets were recorded.

Column	Contents
name, county, subCounty	Identity
level, ownership, origin	primary/secondary, public/private/faith, catalogue or field
latitude, longitude	Decimal degrees
enrollment, teachers, classrooms, toilets, desks	Counts as entered
learnersPerTeacher, learnersPerClassroom, learnersPerToilet	Computed ratios
textbooksPerPupil	Core books per learner
electricity, safeWater, internet, library, scienceLab, computerLab, schoolMeals	Facility flags
resourceIndex, band	0–100 and steady, strained, or critical
notes	Free text

6. Running the project

The app needs Node.js. From the project folder:

```
npm install
npm run dev
```

Then open http://localhost:8080.

A production build is npm run build, then npm run preview. Typecheck is npm run typecheck. The dev server is Vite, through scripts/with-app-env.mjs, on port 8080.

Stack

- TanStack Start and React, with file routes in src/routes.
- Tailwind CSS v4. The palette is a warm paper background, near-black ink, and one forest green. Band colours are the only extra hues, and they are reserved for status.
- Leaflet, loaded only in the browser, with CARTO Positron tiles. Pins are div icons so they match the band colours.
- Zustand with the persist middleware for local edits.
- Recharts for the county bar chart. The chart respects reduced motion.
- Figtree for interface text and Newsreader for the name and headings.

Where the behaviour lives

Path	Role
src/lib/model.ts	Types, planning lines, scoring, filters, county rollup, CSV
src/data/catalogue.ts	The 85-school seed
src/lib/atlas-store.ts	localStorage merge of added schools and overrides
src/components/mapper-app.tsx	Shell, view state, export
src/components/school-map.tsx	Loads Leaflet on the client
src/components/leaflet-map.tsx	Pins, legend, fit, drop-pin

Path	Role
src/components/school-directory.tsx	Search, chips, filters, list
src/components/school-detail.tsx	Profile, shortfalls, edit and revert
src/components/school-form.tsx	Log and update form
src/components/gaps-view.tsx	Priority list
src/components/counties-view.tsx	Chart, table, methodology note
src/routes/index.tsx	The only page

7. Limits

- Profiles are planning composites. Do not quote a figure from this app as an official enrollment, staffing return, or inspection result.
- Coordinates are approximate. They are good enough to separate counties on a national map, not to survey a plot.
- The index compresses twelve different problems into one number. Use the profile and the gap list before deciding what to fund. A school can be critical because of toilets and books while already having power and water.
- Science labs are not part of the primary score. A primary school with no lab is not listed as having that gap.
- Meeting a line caps that part at 1. The index does not reward surplus teachers, books, or desks.
- Map tiles come from a third party. Offline, the list, scores, and export still work; the map does not.
- Field logs and edits never leave the browser they were typed in.

Shule Atlas is a desk tool for seeing where a planning line is missed. The number on the pin is a summary. The sentence under it is the thing to act on.